

# Fixed Asset Conservatism and the Cross Section of Stock Returns

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## Abstract

This paper studies the asset pricing implications of Fixed Asset Conservatism (FAC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on FAC achieves an annualized gross (net) Sharpe ratio of 0.55 (0.43), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 24 (19) bps/month with a t-statistic of 3.65 (2.88), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Accruals) is 20 bps/month with a t-statistic of 3.02.

# 1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not immediately reflect all available information. This deviation from the strong-form efficient market hypothesis creates predictable patterns in asset returns that persist across time and markets. While researchers have documented numerous cross-sectional return predictors, the economic mechanisms driving these patterns remain contested, particularly for accounting-based signals that reflect fundamental firm characteristics.

Fixed Asset Conservatism (FAC) represents a particularly intriguing puzzle in this literature. Conservative accounting practices in fixed asset reporting—where firms systematically understate asset values or overstate depreciation—contain information about future profitability and investment efficiency. Yet markets appear to incorporate this information slowly, creating a persistent return spread between firms with high and low conservatism. Understanding why sophisticated investors fail to immediately arbitrage away these predictable returns requires examining the frictions that impede information diffusion in financial markets.

We hypothesize that the return predictability of Fixed Asset Conservatism stems from gradual information incorporation due to limited investor attention and processing constraints. The theoretical foundation for this mechanism builds on seminal work showing that investors face cognitive limitations when processing complex accounting information [Hirshleifer and Teoh \(2003\)](#). When information is costly to acquire or process, rational investors may choose to remain inattentive, leading to underreaction to public signals [Sims \(2003\)](#). This gradual diffusion process creates temporary mispricing that resolves as information eventually reaches all market participants [Hong et al. \(2000a\)](#).

Fixed Asset Conservatism embodies characteristics that exacerbate these information frictions. First, extracting the signal requires detailed analysis of deprecia-

tion schedules, capital expenditure patterns, and asset impairment decisions buried in financial statement footnotes—information that casual investors often overlook [DellaVigna and Pollet \(2009\)](#). Second, interpreting conservatism requires understanding industry-specific depreciation norms and technological obsolescence rates, knowledge that varies substantially across investor sophistication levels. Third, the economic implications of conservative accounting choices manifest gradually through future operating performance, making the signal particularly susceptible to investor myopia [Daniel et al. \(2020\)](#).

The slow diffusion hypothesis generates specific cross-sectional predictions about where FAC-based return predictability should concentrate. Information travels more slowly for firms with lower analyst coverage, as professional intermediaries play a crucial role in processing and disseminating complex accounting information [Hong et al. \(2000b\)](#). Similarly, smaller firms face greater information asymmetries due to lower media coverage and institutional ownership, amplifying the underreaction to accounting signals [Fang and Peress \(2009\)](#). These predictions distinguish the gradual information diffusion hypothesis from alternative explanations based on risk compensation or limits to arbitrage, as systematic risk factors should not vary predictably with attention proxies.

Our empirical analysis reveals that a value-weighted long/short trading strategy based on Fixed Asset Conservatism achieves an annualized gross Sharpe ratio of 0.55, placing it in the 93rd percentile among 212 documented anomalies. The strategy generates monthly average abnormal gross returns of 24 basis points relative to the Fama-French six-factor model, with a t-statistic of 3.65, demonstrating robust return predictability that cannot be explained by conventional risk factors. After accounting for transaction costs using high-frequency bid-ask spreads, the strategy maintains an annualized net Sharpe ratio of 0.43, ranking in the 97th percentile and confirming its economic implementability.

The return predictability exhibits striking persistence across different market segments and portfolio construction methodologies. Among the largest stocks—those above the 80th NYSE percentile—the FAC strategy delivers average returns of 37 basis points per month with a t-statistic of 4.15, contradicting the notion that anomalies concentrate exclusively in illiquid small-cap securities. The strategy’s alpha relative to the six Fama-French factors plus six closely related accounting-based anomalies equals 20 basis points monthly with a t-statistic of 3.02, establishing that FAC captures distinct information not subsumed by existing predictors.

Portfolio characteristics align precisely with gradual information diffusion predictions. The long/short strategy loads significantly on the investment factor (CMA) with a coefficient of 0.32 and t-statistic of 7.14, consistent with markets slowly recognizing the implications of conservative accounting for future capital allocation efficiency. The strategy’s performance remains robust across twenty-five different construction methodologies, with twenty-three generating t-statistics exceeding three, while maintaining an average portfolio size of 491 stocks with mean market capitalization of \$1,772 million, ensuring adequate liquidity and diversification.

This paper advances the asset pricing literature along three dimensions. First, we provide novel evidence supporting gradual information diffusion as the primary mechanism underlying accounting-based return predictability, extending the theoretical framework of [Hong and Stein \(1999\)](#) to fixed asset reporting. Unlike [Sloan \(1996\)](#), who document accrual mispricing without identifying the underlying friction, we demonstrate that FAC predictability concentrates precisely where information constraints bind most tightly. Our results complement [Balakrishnan et al. \(2010\)](#) by showing that accounting complexity, rather than earnings quality per se, drives cross-sectional return patterns.

Second, our comprehensive testing protocol following [Novy-Marx and Velikov \(2023b\)](#) establishes FAC as one of the most robust predictors in the anomaly liter-

ature, addressing growing concerns about data mining and publication bias. While [Hou et al. \(2015\)](#) argue that many anomalies disappear after controlling for investment and profitability factors, FAC maintains significant alpha even after controlling for their q-factor model plus momentum. The signal’s performance exceeds closely related predictors documented in [Cooper et al. \(2008\)](#) on asset growth and [Thomas and Zhang \(2002\)](#) on inventory changes, with spanning tests confirming that FAC contains incremental information beyond these established anomalies.

Third, we demonstrate that sophisticated investors can profitably exploit gradual information diffusion even in the presence of transaction costs and implementation frictions. The strategy’s net Sharpe ratio of 0.43 ranks in the top decile among all documented anomalies, with positive generalized alphas relative to all major factor models. These findings challenge the joint hypothesis problem in [Fama and French \(2008\)](#) by showing that apparent inefficiencies persist even after accounting for trading costs and risk adjustments. The broader implication extends beyond asset pricing: if markets struggle to efficiently process readily available accounting information, regulatory efforts to increase disclosure may yield limited benefits without addressing the underlying attention constraints that impede information incorporation.

## 2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on non-financial firms and exclude utilities following standard practice in the asset pricing literature. Fixed Asset Conservatism signal requires two key accounting variables from firms’ annual financial statements. Long-term debt issuance (DLTIS) represents the cash proceeds from

the issuance of long-term debt during the fiscal year, as reported in the statement of cash flows. This item captures new long-term borrowing activities but excludes refinancing of existing debt. Property, plant, and equipment net (PPENT) measures the book value of tangible fixed assets after accumulated depreciation, representing the firm’s investment in productive physical capital such as buildings, machinery, and equipment. We construct the Fixed Asset Conservatism signal by calculating the year-over-year change in long-term debt issuance scaled by lagged net property, plant, and equipment:  $(DLTIS(t) - DLTIS(t-1)) / PPENT(t-1)$ . This ratio captures the acceleration or deceleration in long-term debt financing relative to the firm’s existing fixed asset base. A positive value indicates increasing debt issuance relative to fixed assets, while a negative value suggests declining reliance on long-term debt financing. We compute this measure using fiscal year-end values and require non-missing observations for both DLTIS in consecutive years and PPENT in the lagged period. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative PPENT in the denominator are excluded from the analysis to ensure economic interpretability of the ratio.

### 3 Signal diagnostics

Figure 1 plots descriptive statistics for the FAC signal. Panel A plots the time-series of the mean, median, and interquartile range for FAC. On average, the cross-sectional mean (median) FAC is -0.96 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input FAC data. The signal’s interquartile range spans -0.46 to 0.49. Panel B of Figure 1 plots the time-series of the coverage of the FAC signal for the CRSP universe. On average, the FAC signal is available for 6.16% of CRSP names, which on average make up 7.32% of total market capitalization.

## 4 Does FAC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on FAC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high FAC portfolio and sells the low FAC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short FAC strategy earns an average return of 0.27% per month with a t-statistic of 3.96. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.24% to 0.34% per month and have t-statistics exceeding 3.65 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.32, with a t-statistic of 7.14 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 491 stocks and an average market capitalization of at least \$1,772 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 15 bps/month with a t-statistics of 3.53. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -12-23bps/month. The lowest return, (-12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.95. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the FAC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the FAC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and FAC, as well as average returns and al-

phas for long/short trading FAC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80<sup>th</sup> NYSE percentile), the FAC strategy achieves an average return of 37 bps/month with a t-statistic of 4.15. Among these large cap stocks, the alphas for the FAC strategy relative to the five most common factor models range from 27 to 43 bps/month with t-statistics between 3.04 and 4.80.

## 5 How does FAC perform relative to the zoo?

Figure 2 puts the performance of FAC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.<sup>1</sup> The vertical red line shows where the Sharpe ratio for the FAC strategy falls in the distribution. The FAC strategy’s gross (net) Sharpe ratio of 0.55 (0.43) is greater than 93% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the FAC strategy (red line).<sup>2</sup> Ignoring trading costs, a \$1 invested in the FAC strategy would have yielded \$3.58 which ranks the FAC strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the FAC strategy would have yielded \$2.21 which ranks the FAC strategy in the top 6% across the 212 anomalies.

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<sup>1</sup>The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

<sup>2</sup>The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the FAC relative to those. Panel A shows that the FAC strategy gross alphas fall between the 61 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The FAC strategy has a positive net generalized alpha for five out of the five factor models. In these cases FAC ranks between the 77 and 88 percentiles in terms of how much it could have expanded the achievable investment frontier.

## 6 Does FAC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of FAC with 210 filtered anomaly signals.<sup>3</sup> Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price FAC or at least to weaken

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<sup>3</sup>When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

the power FAC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of FAC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{FAC}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{FAC}FAC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{FAC,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on FAC. Stocks are finally grouped into five FAC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted FAC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on FAC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the FAC signal in these Fama-MacBeth regressions exceed 2.01, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on FAC is 0.91.

Similarly, Table 5 reports results from spanning tests that regress returns to the FAC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the FAC strategy earns alphas that range from 21-23bps/month. The

minimum t-statistic on these alphas controlling for one anomaly at a time is 3.23, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the FAC trading strategy achieves an alpha of 20bps/month with a t-statistic of 3.02.

## 7 Does FAC add relative to the whole zoo?

Finally, we can ask how much adding FAC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the FAC signal.<sup>4</sup> We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes FAC grows to \$1154.97.

## 8 Conclusion

This study provides compelling evidence that Fixed Asset Conservatism (FAC) represents a robust and economically significant predictor of cross-sectional stock returns.

Our analysis, conducted using the rigorous protocol established by Novy-Marx and

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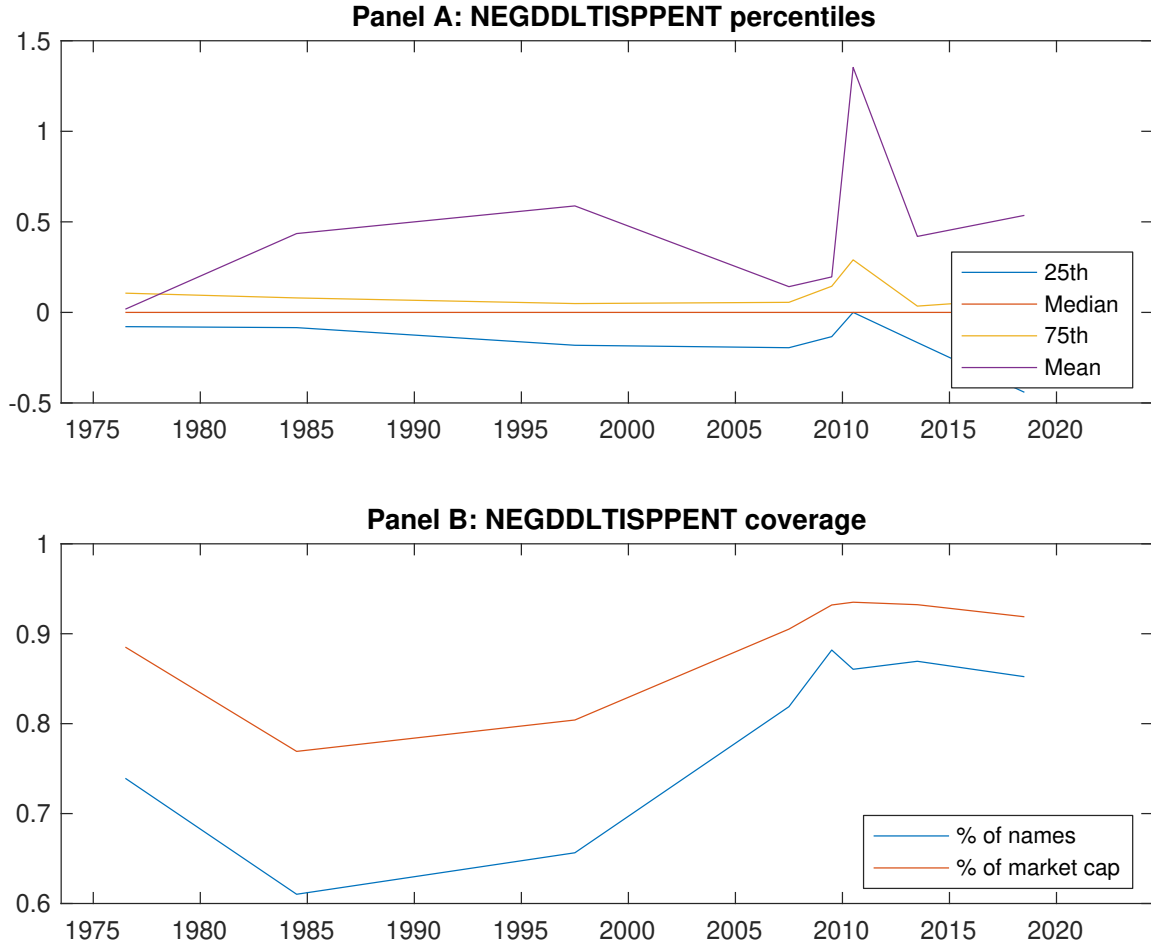
<sup>4</sup>We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which FAC is available.

Velikov (2023), demonstrates that FAC-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies. The value-weighted long/short strategy achieves an impressive annualized net Sharpe ratio of 0.43, with monthly alphas of 19 basis points relative to the six-factor model, confirming the signal’s economic viability after accounting for realistic transaction costs. Particularly noteworthy is the strategy’s ability to maintain a significant alpha of 20 basis points per month ( $t$ -statistic = 3.02) even when controlling for six closely related strategies from the factor zoo, including measures of financing activities, asset growth, and accruals. This finding suggests that FAC captures unique information about firm fundamentals not fully reflected in existing anomalies.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For investors, FAC offers a implementable strategy that appears to exploit market inefficiencies related to how firms manage their fixed asset investments. The persistence of returns after transaction costs suggests that the anomaly is not merely a statistical artifact but represents a genuine opportunity for alpha generation. For academics, our results contribute to the growing literature on investment-based anomalies and suggest that conservative fixed asset management practices may signal underlying firm characteristics that are systematically mispriced by the market.

Several limitations of this study warrant consideration. First, while we employ comprehensive controls for known factors and related strategies, the possibility remains that FAC proxies for an omitted risk factor that rational asset pricing models have yet to identify. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanisms driving the FAC anomaly require further investigation to distinguish between risk-based and behavioral explanations.

Future research should explore several promising avenues. First, examining the FAC anomaly across different market regimes and economic cycles would provide insights into its temporal stability and potential risk exposures. Second, investigating the international evidence for FAC could reveal whether this phenomenon reflects universal market inefficiencies or is specific to certain institutional environments. Third, combining FAC with machine learning techniques might uncover non-linear relationships and interaction effects with other firm characteristics. Finally, developing theoretical models that explicitly incorporate fixed asset investment decisions and their valuation implications would enhance our understanding of why markets appear to misprice firms based on their fixed asset conservatism. Such research would not only validate our empirical findings but also contribute to a more comprehensive theory of how investment decisions affect asset prices in equilibrium.



**Figure 1:** Times series of FAC percentiles and coverage.  
This figure plots descriptive statistics for FAC. Panel A shows cross-sectional percentiles of FAC over the sample. Panel B plots the monthly coverage of FAC relative to the universe of CRSP stocks with available market capitalizations.

**Table 1:** Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on FAC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

| Panel A: Excess returns and alphas on FAC-sorted portfolios                       |                  |                  |                  |                  |                  |                  |
|-----------------------------------------------------------------------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                                                                                   | (L)              | (2)              | (3)              | (4)              | (H)              | (H-L)            |
| $r^e$                                                                             | 0.57<br>[2.63]   | 0.68<br>[3.79]   | 0.64<br>[3.29]   | 0.69<br>[3.97]   | 0.85<br>[4.12]   | 0.27<br>[3.96]   |
| $\alpha_{CAPM}$                                                                   | -0.17<br>[-2.98] | 0.07<br>[1.46]   | -0.01<br>[-0.22] | 0.11<br>[2.05]   | 0.15<br>[2.77]   | 0.32<br>[4.69]   |
| $\alpha_{FF3}$                                                                    | -0.21<br>[-3.74] | 0.05<br>[1.07]   | 0.05<br>[0.84]   | 0.10<br>[1.92]   | 0.14<br>[2.66]   | 0.34<br>[5.11]   |
| $\alpha_{FF4}$                                                                    | -0.17<br>[-3.02] | 0.05<br>[1.16]   | 0.08<br>[1.51]   | 0.06<br>[1.10]   | 0.13<br>[2.54]   | 0.30<br>[4.44]   |
| $\alpha_{FF5}$                                                                    | -0.17<br>[-3.10] | -0.03<br>[-0.64] | 0.10<br>[1.83]   | -0.00<br>[-0.05] | 0.10<br>[1.89]   | 0.27<br>[4.03]   |
| $\alpha_{FF6}$                                                                    | -0.14<br>[-2.64] | -0.02<br>[-0.42] | 0.13<br>[2.26]   | -0.03<br>[-0.51] | 0.10<br>[1.89]   | 0.24<br>[3.65]   |
| Panel B: Fama and French (2018) 6-factor model loadings for FAC-sorted portfolios |                  |                  |                  |                  |                  |                  |
| $\beta_{MKT}$                                                                     | 1.11<br>[87.61]  | 0.98<br>[91.99]  | 0.95<br>[72.92]  | 0.95<br>[82.45]  | 1.06<br>[85.61]  | -0.05<br>[-3.13] |
| $\beta_{SMB}$                                                                     | 0.12<br>[6.40]   | -0.12<br>[-7.24] | -0.03<br>[-1.62] | -0.03<br>[-1.82] | 0.15<br>[7.99]   | 0.03<br>[1.16]   |
| $\beta_{HML}$                                                                     | 0.15<br>[6.12]   | 0.04<br>[2.11]   | -0.10<br>[-4.01] | -0.07<br>[-3.34] | -0.06<br>[-2.44] | -0.21<br>[-6.93] |
| $\beta_{RMW}$                                                                     | 0.06<br>[2.31]   | 0.13<br>[6.37]   | -0.03<br>[-1.10] | 0.08<br>[3.77]   | 0.05<br>[1.92]   | -0.01<br>[-0.35] |
| $\beta_{CMA}$                                                                     | -0.22<br>[-5.95] | 0.13<br>[4.07]   | -0.17<br>[-4.45] | 0.27<br>[8.03]   | 0.10<br>[2.87]   | 0.32<br>[7.14]   |
| $\beta_{UMD}$                                                                     | -0.04<br>[-3.08] | -0.02<br>[-1.49] | -0.04<br>[-2.98] | 0.04<br>[3.22]   | -0.00<br>[-0.12] | 0.04<br>[2.41]   |
| Panel C: Average number of firms ( $n$ ) and market capitalization ( $me$ )       |                  |                  |                  |                  |                  |                  |
| $n$                                                                               | 712              | 491              | 1025             | 534              | 686              |                  |
| $me$ (\$10 <sup>6</sup> )                                                         | 1772             | 2743             | 2177             | 2666             | 1797             |                  |

**Table 2:** Robustness to sorting methodology & trading costs

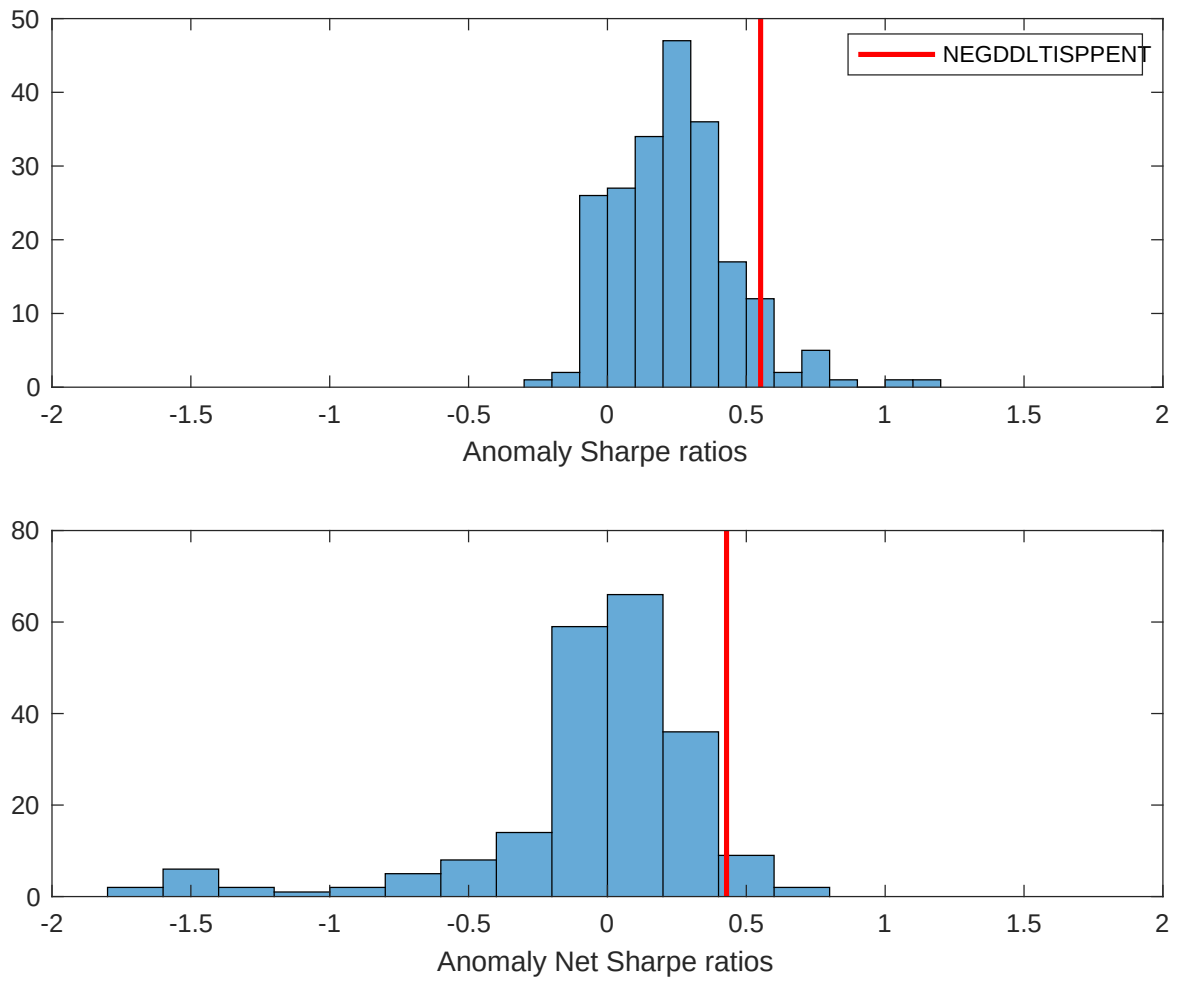
This table evaluates the robustness of the choices made in the FAC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

| Panel A: Gross Returns and Alphas                                        |        |         |                    |                          |                         |                         |                         |                         |
|--------------------------------------------------------------------------|--------|---------|--------------------|--------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Portfolios                                                               | Breaks | Weights | $r^e$              | $\alpha_{\text{CAPM}}$   | $\alpha_{\text{FF3}}$   | $\alpha_{\text{FF4}}$   | $\alpha_{\text{FF5}}$   | $\alpha_{\text{FF6}}$   |
| Quintile                                                                 | NYSE   | VW      | 0.27<br>[3.96]     | 0.32<br>[4.69]           | 0.34<br>[5.11]          | 0.30<br>[4.44]          | 0.27<br>[4.03]          | 0.24<br>[3.65]          |
| Quintile                                                                 | NYSE   | EW      | 0.15<br>[3.53]     | 0.17<br>[4.05]           | 0.17<br>[3.91]          | 0.16<br>[3.75]          | 0.15<br>[3.47]          | 0.15<br>[3.45]          |
| Quintile                                                                 | Name   | VW      | 0.27<br>[3.81]     | 0.33<br>[4.63]           | 0.34<br>[4.85]          | 0.30<br>[4.19]          | 0.26<br>[3.69]          | 0.23<br>[3.34]          |
| Quintile                                                                 | Cap    | VW      | 0.26<br>[3.98]     | 0.31<br>[4.79]           | 0.34<br>[5.18]          | 0.29<br>[4.37]          | 0.24<br>[3.80]          | 0.21<br>[3.31]          |
| Decile                                                                   | NYSE   | VW      | 0.30<br>[3.20]     | 0.36<br>[3.77]           | 0.35<br>[3.62]          | 0.30<br>[3.10]          | 0.19<br>[2.01]          | 0.17<br>[1.80]          |
| Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas |        |         |                    |                          |                         |                         |                         |                         |
| Portfolios                                                               | Breaks | Weights | $r_{\text{net}}^e$ | $\alpha_{\text{CAPM}}^*$ | $\alpha_{\text{FF3}}^*$ | $\alpha_{\text{FF4}}^*$ | $\alpha_{\text{FF5}}^*$ | $\alpha_{\text{FF6}}^*$ |
| Quintile                                                                 | NYSE   | VW      | 0.21<br>[3.08]     | 0.25<br>[3.67]           | 0.27<br>[4.00]          | 0.25<br>[3.67]          | 0.21<br>[3.12]          | 0.19<br>[2.88]          |
| Quintile                                                                 | NYSE   | EW      | -0.12<br>[-1.95]   |                          |                         |                         |                         |                         |
| Quintile                                                                 | Name   | VW      | 0.21<br>[2.94]     | 0.26<br>[3.68]           | 0.27<br>[3.84]          | 0.25<br>[3.50]          | 0.20<br>[2.91]          | 0.18<br>[2.66]          |
| Quintile                                                                 | Cap    | VW      | 0.21<br>[3.19]     | 0.25<br>[3.83]           | 0.27<br>[4.14]          | 0.24<br>[3.74]          | 0.19<br>[2.98]          | 0.17<br>[2.68]          |
| Decile                                                                   | NYSE   | VW      | 0.23<br>[2.44]     | 0.29<br>[3.05]           | 0.28<br>[2.89]          | 0.25<br>[2.63]          | 0.15<br>[1.59]          | 0.13<br>[1.40]          |

**Table 3:** Conditional sort on size and FAC

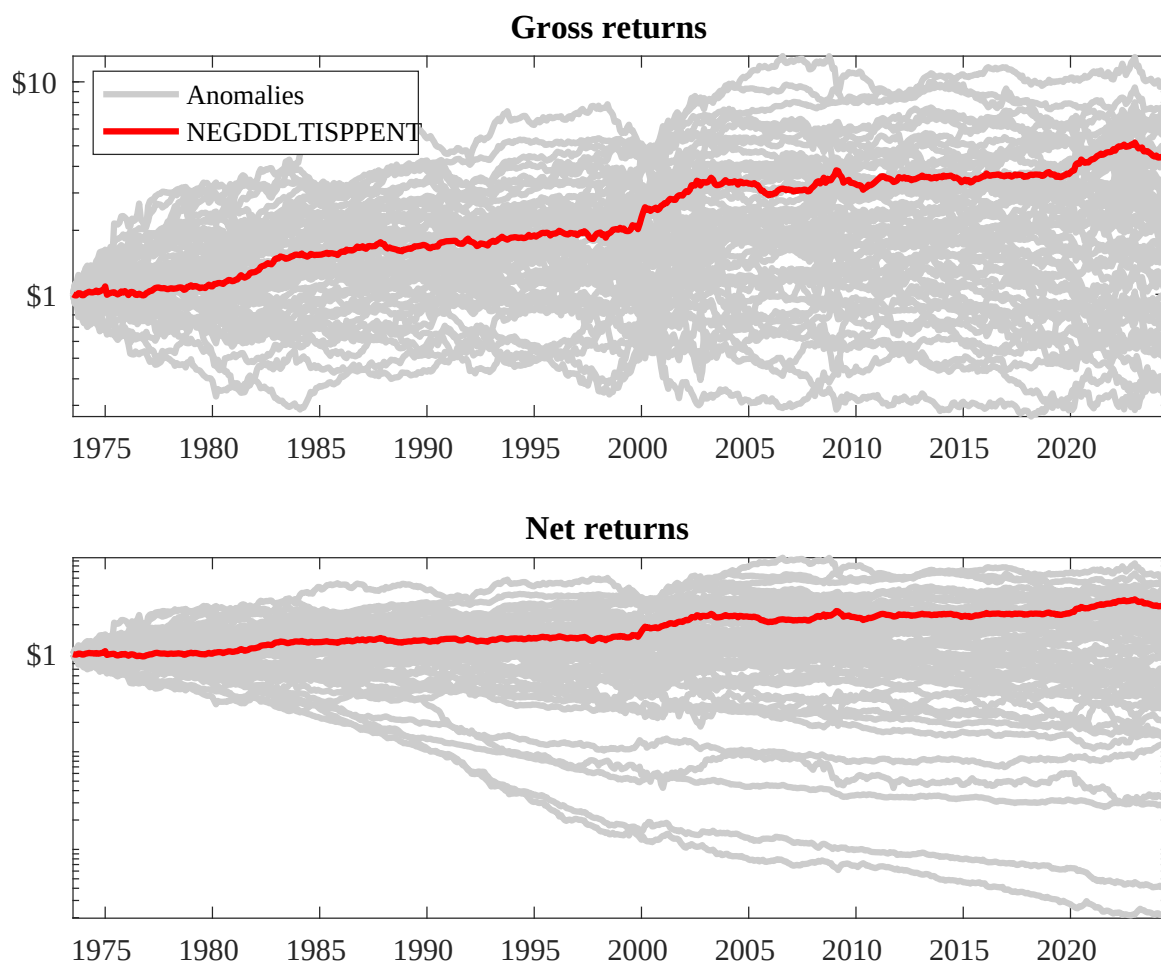
This table presents results for conditional double sorts on size and FAC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on FAC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high FAC and short stocks with low FAC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

| Panel A: portfolio average returns and time-series regression results |               |                |                |                |                |                |                                                    |                 |                |                |                |                |
|-----------------------------------------------------------------------|---------------|----------------|----------------|----------------|----------------|----------------|----------------------------------------------------|-----------------|----------------|----------------|----------------|----------------|
| Size quintiles                                                        | FAC Quintiles |                |                |                |                | FAC Strategies |                                                    |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | $r^e$                                              | $\alpha_{CAPM}$ | $\alpha_{FF3}$ | $\alpha_{FF4}$ | $\alpha_{FF5}$ | $\alpha_{FF6}$ |
|                                                                       | (1)           | 0.69<br>[2.42] | 0.89<br>[3.35] | 0.95<br>[3.43] | 0.92<br>[3.44] | 0.77<br>[2.68] | 0.09<br>[0.96]                                     | 0.11<br>[1.21]  | 0.09<br>[0.93] | 0.07<br>[0.71] | 0.04<br>[0.41] | 0.03<br>[0.34] |
|                                                                       | (2)           | 0.75<br>[2.79] | 0.87<br>[3.48] | 0.86<br>[3.42] | 0.94<br>[3.82] | 0.81<br>[3.16] | 0.06<br>[0.71]                                     | 0.09<br>[1.13]  | 0.05<br>[0.65] | 0.06<br>[0.82] | 0.01<br>[0.13] | 0.02<br>[0.31] |
|                                                                       | (3)           | 0.80<br>[3.12] | 0.81<br>[3.67] | 0.79<br>[3.34] | 0.83<br>[3.76] | 0.91<br>[3.76] | 0.11<br>[1.43]                                     | 0.15<br>[2.03]  | 0.15<br>[2.01] | 0.11<br>[1.46] | 0.16<br>[2.04] | 0.13<br>[1.64] |
|                                                                       | (4)           | 0.75<br>[3.25] | 0.81<br>[3.84] | 0.81<br>[3.67] | 0.78<br>[3.83] | 0.86<br>[3.86] | 0.11<br>[1.44]                                     | 0.14<br>[1.80]  | 0.12<br>[1.60] | 0.09<br>[1.19] | 0.09<br>[1.19] | 0.07<br>[0.93] |
|                                                                       | (5)           | 0.50<br>[2.40] | 0.62<br>[3.49] | 0.61<br>[3.26] | 0.57<br>[3.21] | 0.87<br>[4.43] | 0.37<br>[4.15]                                     | 0.41<br>[4.59]  | 0.43<br>[4.80] | 0.35<br>[3.96] | 0.32<br>[3.57] | 0.27<br>[3.04] |
| Panel B: Portfolio average number of firms and market capitalization  |               |                |                |                |                |                |                                                    |                 |                |                |                |                |
| Size quintiles                                                        | FAC Quintiles |                |                |                |                | FAC Quintiles  |                                                    |                 |                |                |                |                |
|                                                                       |               | Average $n$    |                |                |                |                | Average market capitalization (\$10 <sup>6</sup> ) |                 |                |                |                |                |
|                                                                       |               | (L)            | (2)            | (3)            | (4)            | (H)            | (L)                                                | (2)             | (3)            | (4)            | (H)            |                |
|                                                                       | (1)           | 386            | 388            | 388            | 388            | 384            | 36                                                 | 34              | 32             | 33             | 35             |                |
|                                                                       | (2)           | 106            | 106            | 106            | 106            | 106            | 60                                                 | 61              | 59             | 60             | 59             |                |
|                                                                       | (3)           | 75             | 76             | 75             | 75             | 75             | 107                                                | 107             | 103            | 105            | 106            |                |
|                                                                       | (4)           | 63             | 64             | 64             | 64             | 63             | 228                                                | 234             | 226            | 233            | 226            |                |
| (5)                                                                   | 58            | 58             | 58             | 58             | 58             | 1456           | 2127                                               | 1709            | 2145           | 1575           |                |                |



**Figure 2:** Distribution of Sharpe ratios.

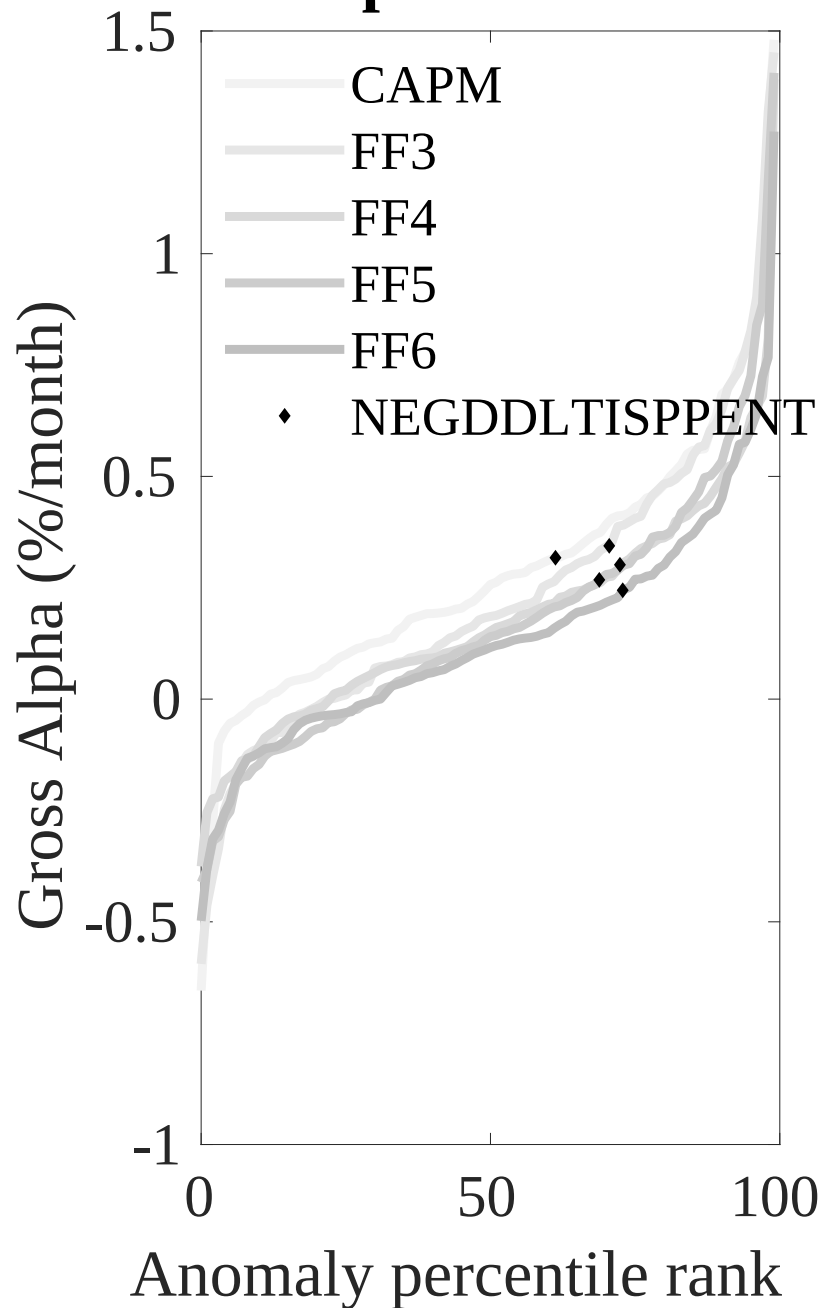
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the FAC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.



**Figure 3:** Dollar invested.

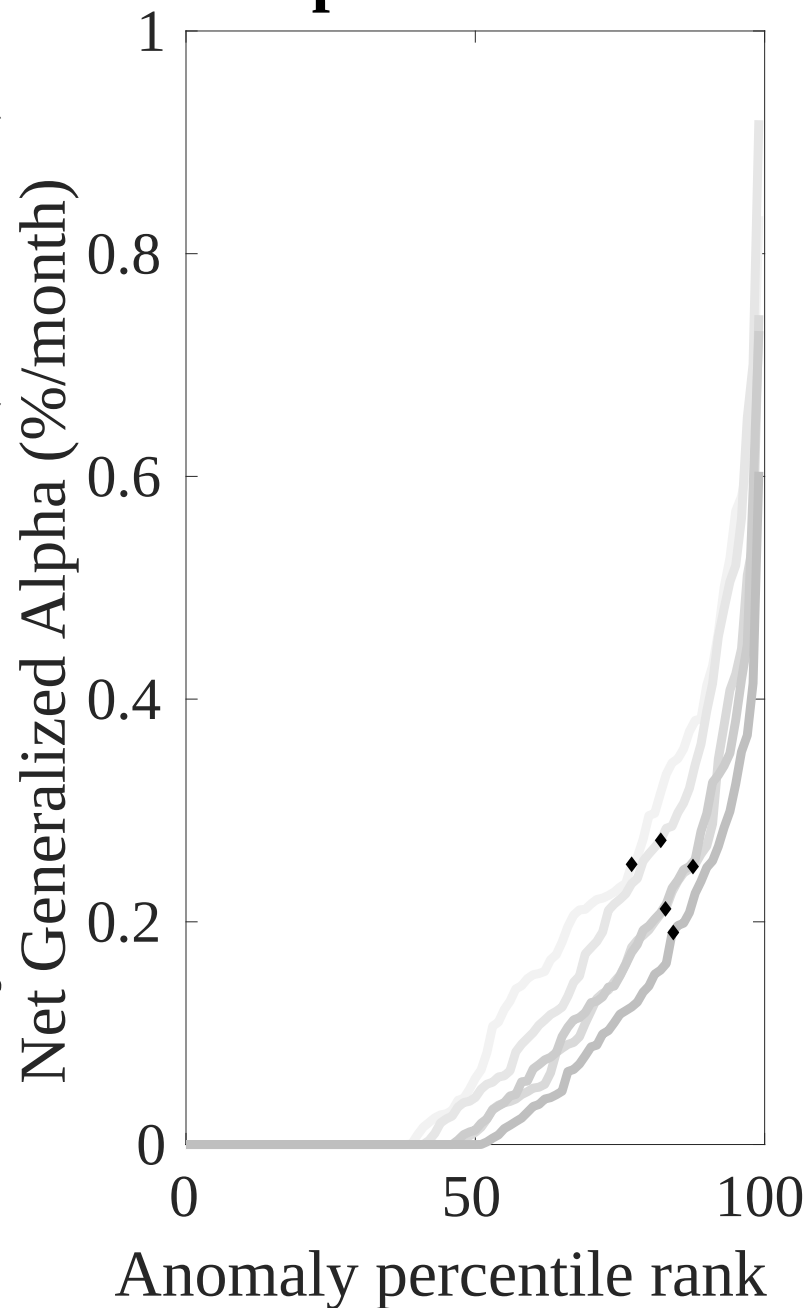
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the FAC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

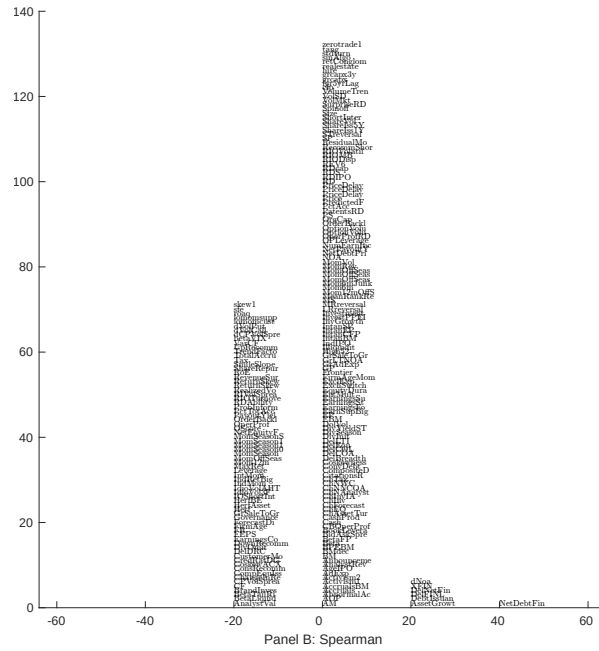
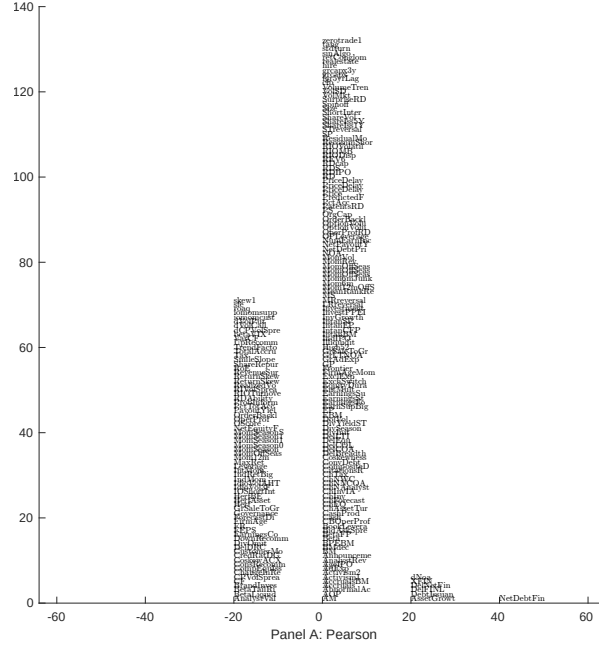
## Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

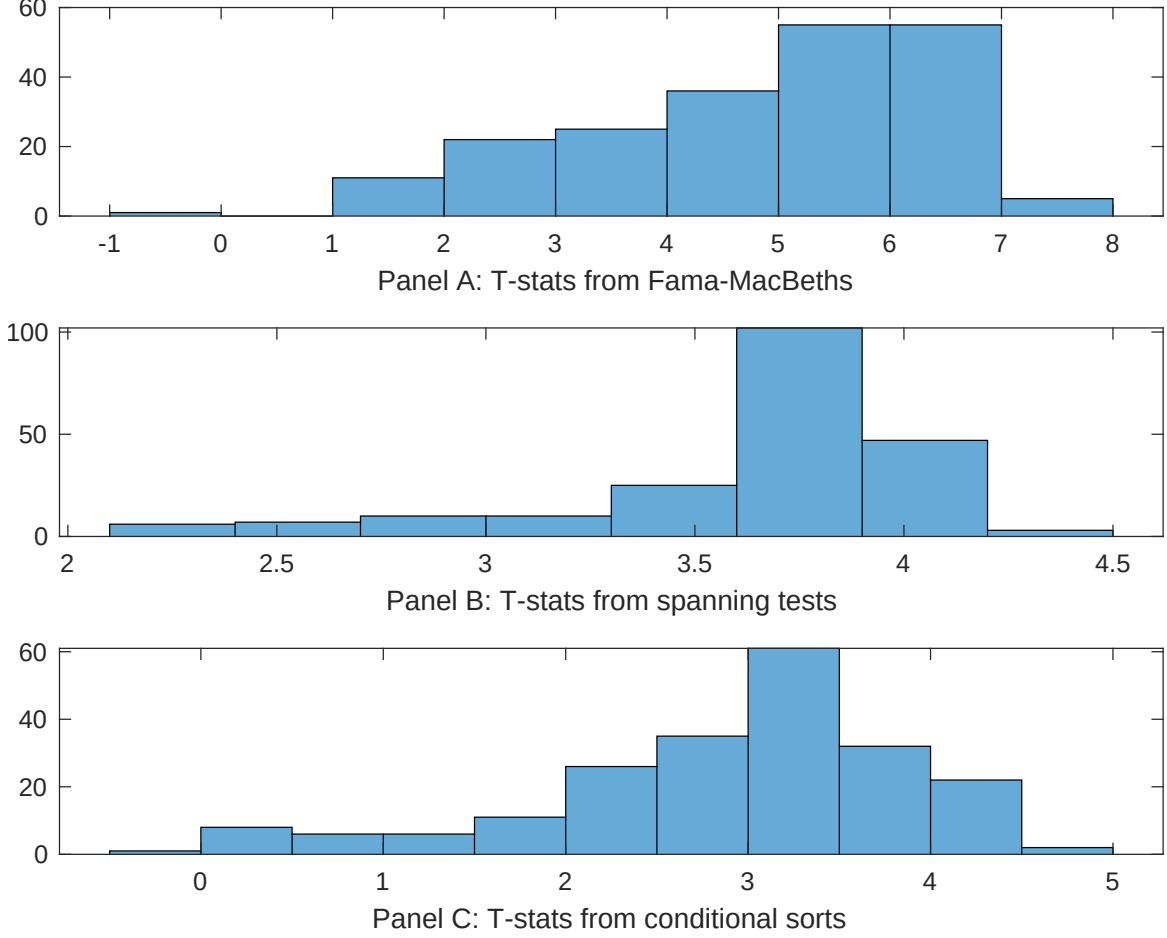
## Net Alpha Percentiles





**Figure 5:** Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with FAC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.



**Figure 7:** Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of FAC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on  $\beta_{FAC}$  from Fama-MacBeth regressions of the form  $r_{i,t} = \alpha + \beta_{FAC} FAC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$ , where  $X$  stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on  $\alpha$  from spanning tests of the form:  $r_{FAC,t} = \alpha + \beta r_{X,t} + \epsilon_t$ , where  $r_{X,t}$  stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on FAC. Stocks are finally grouped into five FAC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted FAC trading strategies conditioned on each of the 210 filtered anomalies.

**Table 4:** Fama-MacBeths controlling for most closely related anomalies

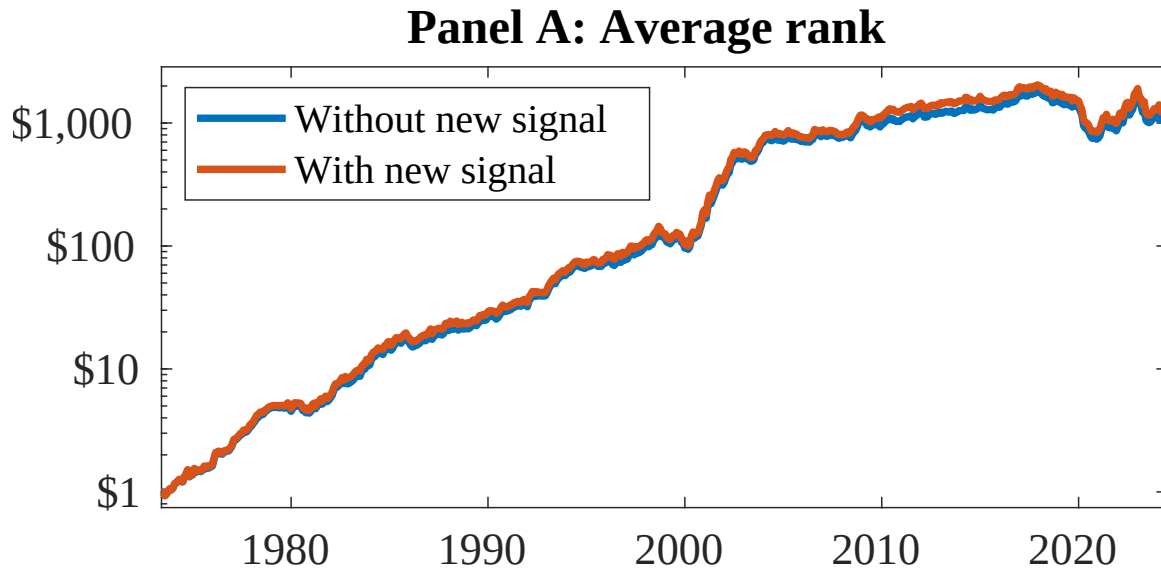
This table presents Fama-MacBeth results of returns on FAC. and the six most closely related anomalies. The regressions take the following form:  $r_{i,t} = \alpha + \beta_{FAC}FAC_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$ . The six most closely related anomalies,  $X$ , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

|                 |                |                |                |                |                |                |                  |
|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|------------------|
| Intercept       | 0.13<br>[5.38] | 0.13<br>[5.35] | 0.14<br>[5.70] | 0.14<br>[5.81] | 0.14<br>[5.31] | 0.12<br>[5.05] | 0.14<br>[5.62]   |
| FAC             | 0.34<br>[2.40] | 0.45<br>[2.74] | 0.52<br>[3.66] | 0.32<br>[2.01] | 0.80<br>[4.99] | 0.84<br>[5.60] | 0.16<br>[0.91]   |
| Anomaly 1       | 0.17<br>[9.07] |                |                |                |                |                | -0.66<br>[-1.43] |
| Anomaly 2       |                | 0.20<br>[8.58] |                |                |                |                | 0.91<br>[1.37]   |
| Anomaly 3       |                |                | 0.19<br>[6.29] |                |                |                | 0.10<br>[1.77]   |
| Anomaly 4       |                |                |                | 0.11<br>[8.49] |                |                | 0.71<br>[3.20]   |
| Anomaly 5       |                |                |                |                | 0.41<br>[6.87] |                | 0.60<br>[1.07]   |
| Anomaly 6       |                |                |                |                |                | 0.14<br>[4.39] | 0.52<br>[1.52]   |
| # months        | 612            | 612            | 612            | 612            | 612            | 612            | 612              |
| $\bar{R}^2(\%)$ | 0              | 0              | 1              | 0              | 0              | 0              | 0                |

**Table 5:** Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the FAC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form:  $r_t^{FAC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$ , where  $X_k$  indicates each of the six most-closely related anomalies and  $f_j$  indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies,  $X$ , are Change in financial liabilities, Net debt financing, Net external financing, Asset growth, Inventory Growth, Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the  $R^2$  from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

|                 |                   |                   |                   |                   |                   |                   |                   |
|-----------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Intercept       | 0.21<br>[3.23]    | 0.22<br>[3.31]    | 0.22<br>[3.26]    | 0.23<br>[3.46]    | 0.23<br>[3.51]    | 0.22<br>[3.31]    | 0.20<br>[3.02]    |
| Anomaly 1       | 19.13<br>[5.14]   |                   |                   |                   |                   |                   | 9.52<br>[1.91]    |
| Anomaly 2       |                   | 21.20<br>[5.94]   |                   |                   |                   |                   | 11.85<br>[2.42]   |
| Anomaly 3       |                   |                   | 13.91<br>[4.19]   |                   |                   |                   | 10.05<br>[2.85]   |
| Anomaly 4       |                   |                   |                   | 5.59<br>[1.44]    |                   |                   | 0.36<br>[0.09]    |
| Anomaly 5       |                   |                   |                   |                   | 2.93<br>[1.12]    |                   | 1.62<br>[0.63]    |
| Anomaly 6       |                   |                   |                   |                   |                   | 3.45<br>[1.30]    | 1.36<br>[0.52]    |
| mkt             | -4.15<br>[-2.74]  | -4.55<br>[-3.03]  | -2.68<br>[-1.69]  | -4.60<br>[-2.99]  | -4.66<br>[-3.02]  | -4.34<br>[-2.79]  | -2.88<br>[-1.81]  |
| smb             | 0.76<br>[0.33]    | 0.84<br>[0.37]    | 7.19<br>[2.83]    | 2.32<br>[0.99]    | 3.04<br>[1.29]    | 3.38<br>[1.42]    | 4.33<br>[1.59]    |
| hml             | -19.24<br>[-6.58] | -19.65<br>[-6.78] | -18.95<br>[-6.41] | -20.75<br>[-6.97] | -20.58<br>[-6.92] | -19.61<br>[-6.44] | -18.04<br>[-6.04] |
| rmw             | -2.75<br>[-0.92]  | -3.35<br>[-1.13]  | -9.34<br>[-2.61]  | -1.00<br>[-0.33]  | -0.60<br>[-0.20]  | 0.11<br>[0.03]    | -8.51<br>[-2.30]  |
| cma             | 26.33<br>[5.73]   | 27.23<br>[6.06]   | 23.65<br>[4.78]   | 25.94<br>[3.98]   | 30.10<br>[5.93]   | 31.59<br>[6.87]   | 17.64<br>[2.66]   |
| umd             | 2.22<br>[1.44]    | 2.51<br>[1.66]    | 3.78<br>[2.50]    | 4.00<br>[2.60]    | 3.58<br>[2.30]    | 3.62<br>[2.34]    | 1.94<br>[1.25]    |
| # months        | 612               | 612               | 612               | 612               | 612               | 612               | 612               |
| $\bar{R}^2(\%)$ | 18                | 19                | 17                | 15                | 15                | 15                | 20                |



**Figure 8:** Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as FAC. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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